AI Data Workspace

Official User Guide & Documentation

Welcome to the **AI Data Workspace**, your secure, LLM-powered sandbox for data analysis, visualization, and manipulation. This guide will walk you through the core features of the platform, from uploading data to utilizing the autonomous Magic Clean agent.

1. Getting Started: Data Ingestion

The workspace supports seamless ingestion of tabular data. The system automatically detects file encodings and uses Magic Byte sniffing to prevent corrupted uploads.

Uploading Files

- Click on the **Ingestion** panel or drag-and-drop your file.
- Supported formats: `.CSV`, `.XLSX`, and `.XLS`.
- Click **Load into Sandbox**. A unique, isolated session is created for your data to ensure privacy and prevent cross-contamination.

Note: If you refresh your browser, your session is automatically persisted and will reload instantly.

2. The Query Sandbox (IDE)

The core of the application is a secure Python execution environment. You can write raw Pandas code to manipulate your dataset.

Writing & Executing Code

Your data is automatically loaded into a Pandas DataFrame named `df`. You can manipulate it directly.

```
# Example: Creating a new column
df['Total_Sales'] = df['Price'] * df['Quantity']
```

Click **Execute Query**. The background worker processes the query in a secure Docker container, updates your data table, and streams any `print()` statements directly to the **Console Output** terminal.

3. Data Visualization

The sandbox fully supports data visualization using both static and interactive libraries.

Creating Interactive Charts (Plotly)

The `plotly.express` library is pre-loaded as `px`. To render a chart, assign it to a variable named `fig`.

```
# Example: Interactive Bar Chart
fig = px.bar(df, x='Category', y='Sales', title='Sales by Category')
```

Your interactive chart will appear below the console. You can hover, pan, zoom, and download it as a PNG.

4. The AI Copilot & Magic Clean

You don't need to be a Pandas expert to use this workspace. The integrated Llama-3 Copilot can write the code for you.

Natural Language Queries

Type what you want to achieve in the Copilot input box (e.g., *"Group by Department and calculate the average salary"*). The AI will generate the exact, error-free Pandas code. Click **Transfer to Editor** → to drop it into your sandbox.

✨ **Pro Tip: The Magic Clean Button**
Dirty data? Click the **Magic Clean** button. The platform will autonomously profile your dataset (checking for missing values and duplicates), send the profile to the AI, and generate a comprehensive cleaning script tailored specifically to your data's issues.

5. Unified Export

Once you have finished cleaning and manipulating your data, you can export it with a single click.

- Navigate to the top right of the **Workspace View**.
- Click **.CSV** or **.XLSX**.
- The server will compile your session data and download the fresh, updated file directly to your machine.

6. Security Architecture (Under the Hood)

This platform is built with enterprise-grade security. When you click Execute, your Python code is not run on the main server. Instead, it is routed through a Redis Message Queue and executed inside a disposable, isolated Linux Docker Container. This guarantees that malicious code cannot access the host machine.